



INVISIBLE ERP V2 · OSHINEI ENTERPRISE

System Overview

Program Flow · Capabilities · Key Strengths

A fully integrated ERP + POS platform for Southeast-Asian business — multi-tenant, with SOX/ICFR-grade internal controls, prepared for a NASDAQ listing.

Executive Summary · Issued 29 June 2026

NestJS (Fastify) API · Next.js Web · Drizzle ORM · PostgreSQL (RLS) · 5 languages

1. Executive Overview

Invisible ERP V2 is an enterprise resource-planning system with point-of-sale (POS) built in, covering every business cycle from sales, procurement and inventory through to the general ledger, tax, payroll and AI-driven analytics. It is designed control-first — internal control is the architectural core — to support audit at international standard.

120+ business modules

Spanning 17 core business cycles, from the shop floor to consolidated financial statements.

168+ internal screens

Plus 10+ customer-portal surfaces, each gated by permission.

77 controls

Mapped to the COSO/ICFR framework, with a generated RCM and SoD matrix.

5 languages

Thai (primary), English, Malay, Vietnamese, Indonesian.

Technology architecture

Component	Technology
Backend API	NestJS 11 (Fastify) — port 8000
Frontend Web	Next.js 15 (App Router) + React 18 + TypeScript
Database	PostgreSQL 16 + Row-Level Security (automatic per-tenant isolation)
ORM	Drizzle ORM
Security	JWT + Scrypt hashing + TOTP MFA (two-factor)
Mobile	PWA — installable as an app on iOS / Android

2. How It's Used — Two Workspaces

The system splits into **two workspaces** switchable from the sidebar, each showing menus filtered to the user's role and permissions.



ERP Workspace

Back office — accounting, finance, procurement, inventory, production, HR, planning and BI for office staff.



POS Workspace

Front of house — sales, returns, table management, kitchen (KDS), shift close, for cashiers and store managers.

Main menu groups (19)

Dashboard POS Sales Delivery & Claims Devices & Payments CRM Inventory Procurement
Production Finance Tax HR & Payroll Planning & BI Loyalty Pricing & Branches Controls
AI Assistant Restaurant Analytics System Settings

Productivity features

- ★ **Favourites:** pin frequently-used menus; synced across devices via the user account.
- **Recents:** auto-tracks the menus you just visited.
- **Command palette (Cmd+K):** keyboard search across every page.
- **Three sign-in modes:** password / PIN (cashiers can open a till shift instantly) / SSO (SAML, OIDC).

3. Core Business-Cycle Flows

3.1 Revenue cycle — Order-to-Cash

- 1 Capture POS order — credit-limit / 90-day-overdue check before the sale is allowed.
- 2 Allocate document number automatically (SALE-/INV-) with a no-duplicate lock.
- 3 Take payment — card / e-wallet / QR via PSP (Stripe, Omise); recorded Pending before the real capture.
- 4 Issue AR invoice and apply receipts on payment (RCP-).
- 5 Collections & dunning — staged ladder (reminder → notice → final → legal), dispatched automatically.
- 6 Auto GL posting — revenue + 7% VAT, balanced by construction.

3.2 Expenditure cycle — Procure-to-Pay

- 1 Raise PR — any employee can request → **approve PO** by amount threshold (maker-checker).
- 2 Vendor blocklist **gate** before a PO is issued, blocking purchases from disallowed vendors.
- 3 Warehouse receives goods (**GR**) — permission segregated from the buyer (anti-fraud).
- 4 Three-way match — PO ↔ receipt ↔ invoice within tolerance.
- 5 Two-step payment — the requester (maker) must differ from the approver (checker), or the system blocks it.

3.3 General Ledger & period close

- 1 Record journal (JE) — debit=credit checked automatically; posts as Draft first.
- 2 Approved by a different person (**GL-05**) — even Admin cannot self-approve; only then does it hit the trial balance.
- 3 Reconcile AR/AP/inventory/assets ↔ GL with two-person sign-off.
- 4 Close the period — locks against back-dated posting; produces TB / P&L / balance sheet / cash-flow statement.

4. What the System Can Do

Business cycle	Key capabilities
POS / front-of-house	Sell / return / refund, table management, kitchen KDS, split bills, gift cards, shift close / Z-report, e-tax invoices.
Finance AR/AP	Invoicing & receipts, collections / dunning, credit limits, advances, bad-debt write-off (maker-checker), multi-currency.
General Ledger	Auto posting, recurring & prepaid journals, cash-flow statement, chart of accounts, period / year-end close.
Inventory	Multi-warehouse stock, stocktake, adjustments, lot / expiry, FIFO / weighted-average costing, auto-replenishment, mobile scan.
Procurement	PR/PO/RFQ, three-way match, supplier scorecards, invoice OCR, supplier portal.
Assets & leases	Asset register, depreciation, revaluation / impairment, maintenance (EAM), IFRS 16 leases.
Tax	VAT, withholding tax (WHT), e-Tax, e-Invoice per Thai standards.
HR & Payroll	Employee register, scheduling, overtime, payroll, employee self-service (ESS), expense approvals.
Production	Bill of materials (BoM), manufacturing orders, MRP, production costing.
CRM & Loyalty	360° customer view, sales pipeline, members / points / tiers, missions / wheels, LINE & SMS campaigns.
Multi-entity / branch	Consolidated statements, intercompany, FX translation, offline POS across branches.
Analytics & AI	Demand forecasting, anomaly detection, AI assistant (19 tools), natural-language analytics, scheduled reports.
Customer portal	Self-service B2B ordering, stock visibility, order tracking, mini-ERP (manage own customers / suppliers).

5. Key Strengths

Compliance-first by design

- **Maker-checker everywhere:** posting / payment / approval needs two people — even Admin cannot approve its own work.
- **Automated segregation of duties (SoD):** 16 conflict rules + 16 single-duty roles designed with zero conflicts each.
- **Immutable audit trail:** activity logs are locked at the database level (trigger), preventing edit / delete.
- **Two-factor MFA:** enforced for sensitive permissions (accounting, payments, user administration).

International accounting rigor

- **Balanced by construction:** every JE forces debit = credit before posting.
- **Control-account guards:** no direct posting to AR/AP/inventory/assets — must flow through the subledger.
- **Subledger ↔ GL tie-out:** all control balances reconciled in a single view.
- **International standards:** IFRS 16 (leases), indirect-method cash-flow statement, multi-currency, consolidation.

Full Thai-business support

- 7% VAT, withholding tax, e-Tax, e-Invoice to Revenue Department standard.
- Business timezone Asia/Bangkok (UTC+7) — everything dated on the true business day.
- Buddhist-era dates, Baht currency, Sarabun font.

Built-in AI & analytics

- **Agentic AI assistant:** calls 19 real tools (not hallucination), permission-gated, with PDPA PII redaction.
- **Accurate forecasting:** computes reorder points, safety stock, and anomaly detection via Z-score.
- **Natural-language analytics:** ask business questions in plain language.

Omnichannel & extensible

- **Omnichannel:** store + web + mobile + marketplace, with sales / stock / points unified.
- **No-code customization:** add fields / objects, build approval workflows, design dashboards.
- **External integration:** webhooks, Zapier, developer API.

Thoroughly tested on every change

Eight automated CI gates cover accounting correctness, transactional integrity, analytics and end-to-end flows — every change must pass before it merges — backed by 27 process narratives, 15 user-manual guides, and a complete SOX/ICFR audit-documentation set.

